

Exp No: 9

ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

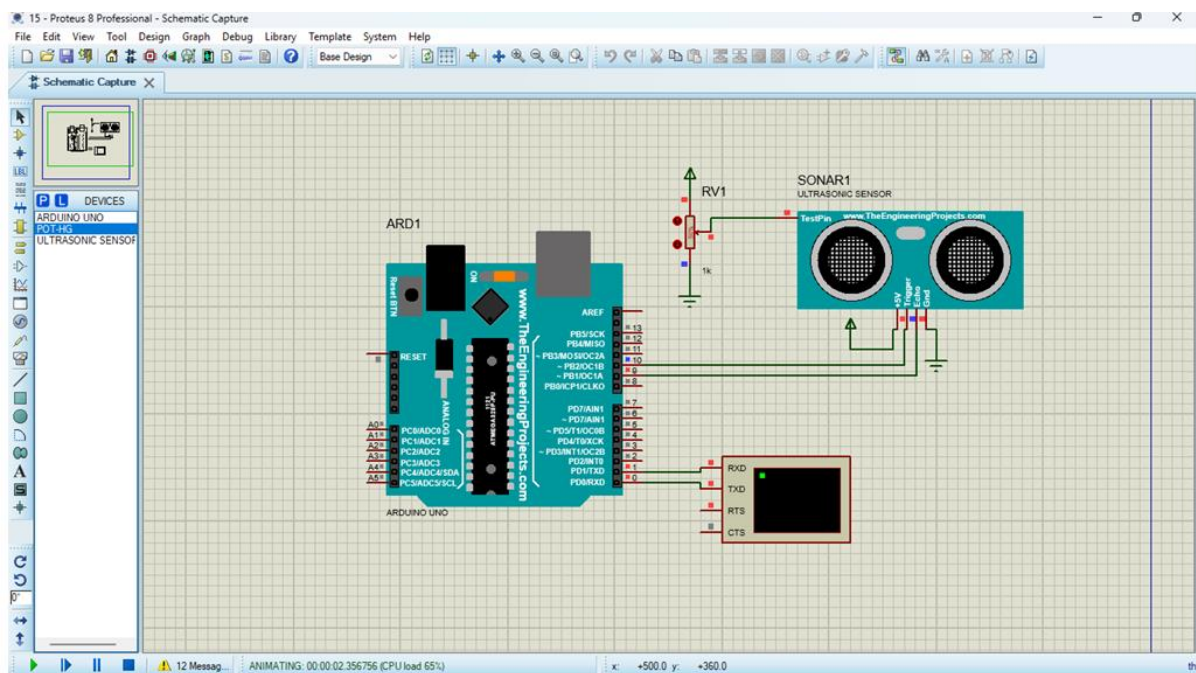
AIM:

To write an Embedded C program for interfacing Ultrasonic Sensor using Arduino Uno and Proteus

SOFTWARE REQUIRED:

- Proteus 8 software.
- Arduino IDE software

CIRCUIT DIAGRAM:



PROGRAM:

// Program Sketch

```
int trig = 10;
```

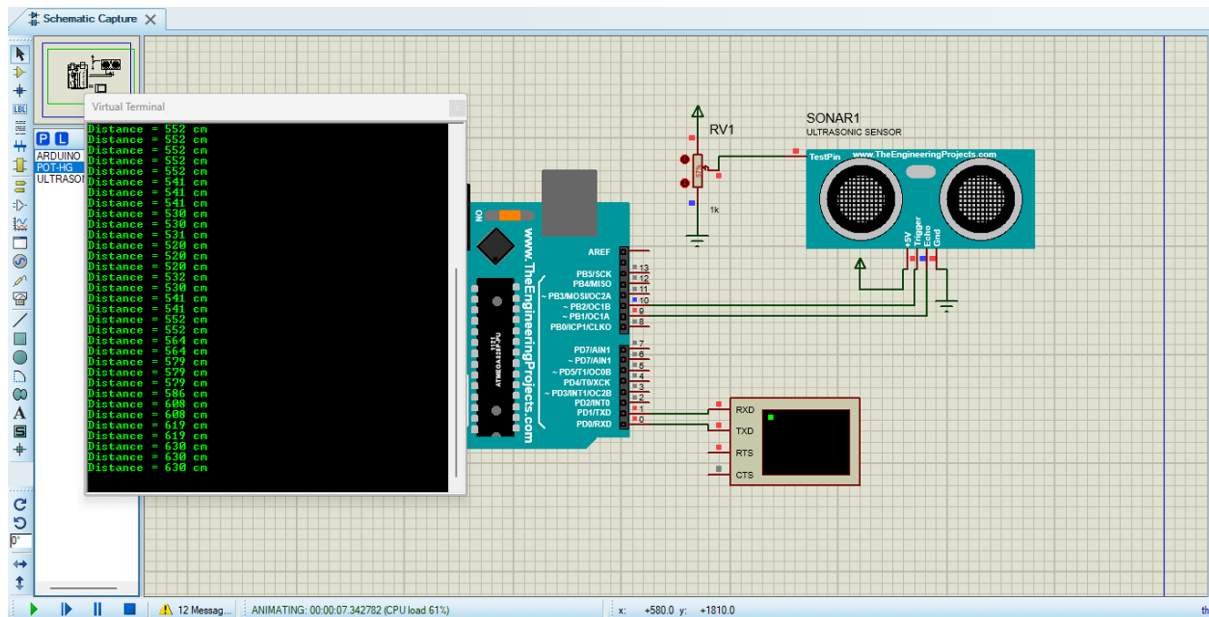
```
int echo = 9;
```

```
long duration;
```

```
int cm;
```

```
void setup() {  
  // put your setup code here, to run once:  
  pinMode(trig, OUTPUT);  
  pinMode(echo, INPUT);  
  Serial.begin(9600);  
}  
  
void loop() { // Sketch to calculate ultrasonic distance  
  digitalWrite(trig, LOW);  
  delayMicroseconds(10);  
  digitalWrite(trig, HIGH);  
  delayMicroseconds(10);  
  digitalWrite(trig, LOW);  
  delayMicroseconds(10);  
  duration = pulseIn(echo, HIGH);  
  cm = (duration/2) * 0.034;  
  // Print Value on Serial Monitor  
  Serial.print(" Distance = ");  
  Serial.print(cm);  
  Serial.println(" cm");  
}
```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.